Ch6.2 Regularization

Thursday, October 15, 2026 9:24 AM

Recap: Battle overfitting

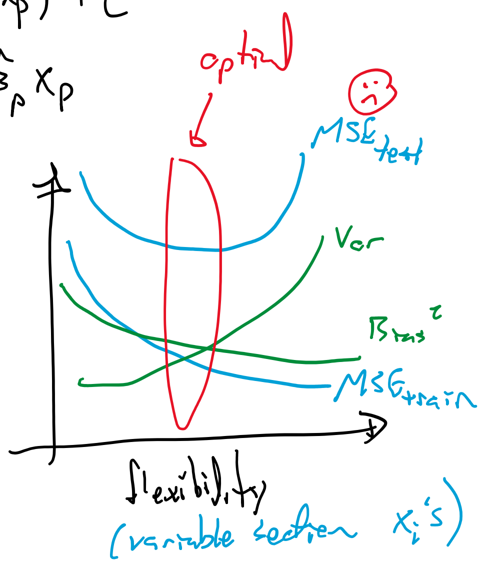
LR $y = f(x) + \varepsilon = (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$

$f(x) \approx \hat{f}(x) = \hat\beta_0 + \hat\beta_1 X_1 + \dots + \hat\beta_p X_p$

2 goals WRT LR

① $p \approx n$, avoid overfitting →

⊛ ② If p largeish (20), hard to interpret $\hat\beta_i$'s.



2 approaches: ① Variable selection by trying many x_i combinations (best, $\underbrace{\text{forw, back}}_{\text{practical}}$)

& assess via

(i) stat metrics (adj R^2, BIC, AIC, C_p)

⊛ (ii) CV

② Regularization: when fit $\hat f$, include variable selection by restricting $\hat\beta_i$'s.

shape of cost function

only shrinkage

(i) Ridge (l^2-penalty)

(ii) LASSO (l^1-penalty)

↳ least absolute shrinkage & $\boxed{\text{selection}}$ operator

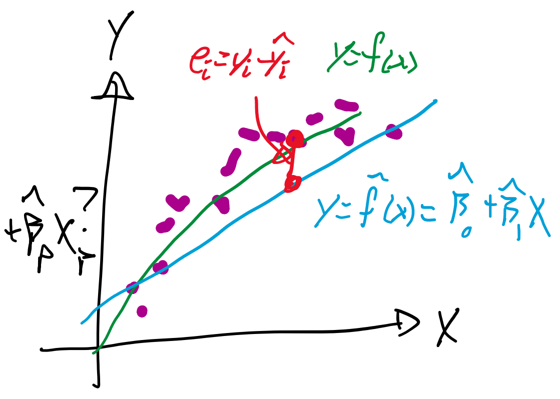
Combo

(iii) Elastic net (lab)

Recap: OLS (ordinary least squares)

$Y = f(x) + \varepsilon = (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$

How to fit $f(x) \approx \hat f(x) = \hat\beta_0 + \hat\beta_1 X_1 + \dots + \hat\beta_p X_p$?



values of $\vec\beta$ at which

$$\hat{\vec\beta} = \underset{\vec\beta}{\boxed{\text{arg}}\ \text{min}}\ RSS = \underset{\vec\beta}{\arg\min} \sum_{i=1}^{n} e_i^2$$

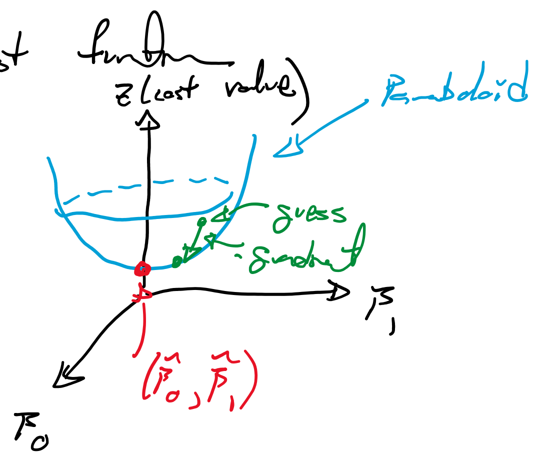
$$= \underset{\vec\beta}{\arg\min} \sum_{i=1}^{n} (y_i - \hat y_i)^2$$

$$= \underset{\vec\beta}{\arg\min} \underbrace{\sum_{i=1}^{n} \Big(y_i - \underbrace{\big(\beta_0 + \sum_{j=1}^{p} \beta_j X_{ji}\big)}_{f(x)}\Big)^2}_{\text{cost function}}$$

(2 ways to optimize:
① Normal eqns
② $\underbrace{\text{Gradient descent}}_{\text{Gen'l ML}}$)

G.D. Plot of cost function

SLR



Details in NN.

Ridge Regression Idea: When OLR overfits, $\hat\beta_i$'s get very big. Force them to stay small (& not overfit).

Standardize (z-score) $z_i = \dfrac{x_i - \bar x_i}{SD(x_i)}$, x_i = orig (mean 0 & SD 1)

Modify cost function

regularization param to tune

β_j gets big, adds to RSS & optimization avoids

shrinkage

$$\hat{\vec\beta}^R_\lambda = \underset{\vec\beta}{\arg\min}\Big(RSS + \underbrace{\lambda \sum_{j=1}^{p} \beta_j^2}_{\vec\beta \text{ penalty}}\Big)$$

$$= \underset{\vec\beta}{\arg\min}\Big(\sum_{i=1}^{n} \big(y_i - (\beta_0 + \sum_{j=1}^{p} \beta_j X_{ji})\big)^2 + \underline{\lambda \sum_{j=1}^{p} \beta_j^2}\Big)$$ (via G.D.)

Warning: OLS gives same $\vec\beta$'s regardless of data scale (need not normalize)

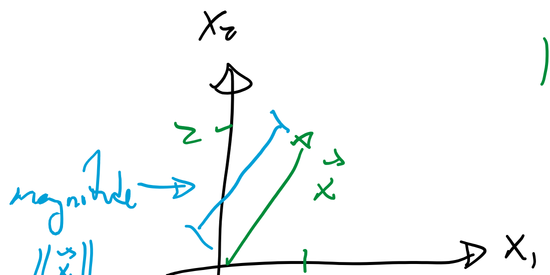
Ridge regression needs normalization else $\hat\beta_i^R$ skew per x_i units (un-normalize after fitting)

Recall: $\sum_{j=1}^{p} \beta_j^2 = \|\vec\beta\|_2^2$, $\vec\beta = \begin{bmatrix}\beta_1\\ \vdots \\ \beta_p\end{bmatrix}$

l^2-norm

Euclidean dist

$\|\vec x\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}$



Tune λ to balance over/under fitting:

$\vec\beta$

CV on λ range!

- $\lambda \approx 0$, OLS ish (overfit)
- * between best
- λ big, $\hat\beta_i$'s super small (underfit)

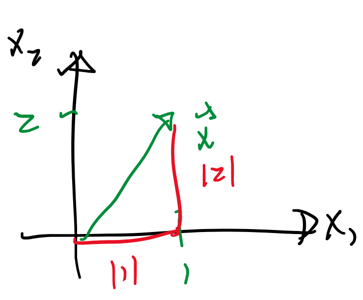
$\hat{\vec\beta}^R_\lambda$

LASSO Same as ridge regr, but $\|\cdot\|_1$ (l^1-norm instead).

$$\hat{\vec\beta}^L_\lambda = \underset{\vec\beta}{\arg\min}\Big(RSS + \lambda \sum_{j=1}^{p} |\beta_j|\Big)$$

(Difference is β_j's get 0ed! cut & shrunk)

Note: $\|\vec x\|_1 = |1| + |2| = 3$ (taxicab or manhattan distance)



Note: Optimization reform. gives LASSO intuition.